Assignment 5

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Question 1

Let

$$S_0 = 6600, \quad S_T = \text{SPX in 1 year}, \quad R = \frac{S_T - S_0}{S_0}$$

The contract pays (in \$m):

$$Payoff = \begin{cases} 9 & R < -0.10 \;\Leftrightarrow\; S_T < 0.9S_0 = 5940 \\ 10(1+R) = 10\dfrac{S_T}{S_0} & -0.10 \le R \le 0.12 \;\Leftrightarrow\; 5940 \le S_T \le 7392 \\ 11.2 & R > 0.12 \;\Leftrightarrow\; S_T > 7392 \end{cases}$$

On the middle region we need a payoff

$$10\frac{S_T}{S_0} \quad \Rightarrow \quad \frac{10{,}000{,}000}{6600} = 1515.15 \text{ \$/index point.}$$

Each SPX futures contract is \$100 per index point, so

$$N = \frac{1515.15}{100} \approx 15.15.$$

Thus, we should be <u>long of N SPX futures (1Y)</u> and <u>invest the PV corresponding to \$10m in T-bills maturing in 1 year</u>.

Then, we need to add the floor and the cap by using options. For that, we can define the strikes

$$K_1 = 5940 \quad (R = -10\%), \qquad K_2 = 7392 \quad (R = +12\%).$$

Then if we choose to long N European puts with $K_1 = 5940$ and a 1-year maturity and to short N European calls with $K_2 = 7392$ and a 1-year maturity, we get a total payoff of :

- If $S_T < K_1$: put ITM, call OTM, futures loss is offset by the put. The payoff is thus: 9 million
- If $K_1 \le S_T \le K_2$: both options OTM so the payoff is what we calculated before which is: $10\dfrac{S_T}{S_0}$ million
- If $S_T > K_2$: call ITM, put OTM, extra futures gain is offset by the short call. The payoff is thus: 11.2 million

Hence the replicating 1-year strategy which matches the pension payoff:

Invest present value of \$10m in T-bills,
Long 15.15 SPX futures,
Long 15.15 puts ($K = 5940,\ T = 1\text{Y}$),
Short 15.15 calls ($K = 7392,\ T = 1\text{Y}$),

Question 2

Question a

We consider three risky assets:

- SP 500 index (US Equity)
- SP World index (Global Equity)
- SP US Treasury Bond index

The risk-free rate is proxied by the 4-week Treasury Bill (DTB4WK).

Question b

Using 10 years of daily data, returns are computed and annualised. Excess returns are obtained by subtracting the risk-free rate.
Annualised expected excess returns:

$$\mu = \begin{pmatrix} 0.1094 \\ 0.0758 \\ -0.0045 \end{pmatrix}$$

Annualised covariance matrix:

$$\Sigma = \begin{pmatrix} 0.0331 & 0.0265 & -0.0011 \\ 0.0265 & 0.0232 & -0.0007 \\ -0.0011 & -0.0007 & 0.0017 \end{pmatrix}$$

Market capitalisations are respectively 60, 60 and 30 trillion USD. Thus, market weights are :

$$w = \begin{pmatrix} 0.4 & 0.4 & 0.2 \end{pmatrix}$$

The parameter δ captures the degree of risk aversion of the representative investor: higher values imply a stronger penalty on variance and therefore more conservative portfolios. According to the lecture slides, using $\delta = 2.5$ together with the covariance matrix and the market weights produces the equilibrium risk premia that characterize the consensus market view. This value yields expected equity risk premia consistent with empirically observed ranges and reflects a moderate level of risk aversion typical of institutional investors. Therefore, we choose

$$\boxed{\delta = 2.5}$$

In the Black–Litterman model, the covariance of the prior distribution is given by $\tau\Sigma$, where Σ is the covariance matrix of realized returns and τ is a scalar capturing the degree of uncertainty surrounding the equilibrium returns. The lecture material emphasizes that smaller values of τ correspond to greater confidence in the market equilibrium, reducing the impact of investor views, while larger values increase the influence of subjective views on the posterior returns. In practice, the literature suggests using values in the range

$$0.025 \leq \tau \leq 0.1,$$

which balances stability against flexibility when incorporating investor opinions.
Here, we choose

$$\boxed{\tau = 0.05}$$

which lies at the midpoint of this recommended interval.

Equilibrium excess returns are then :

$$\pi = \delta\Sigma w = \begin{pmatrix} 0.0591 \\ 0.0494 \\ -0.0010 \end{pmatrix}$$

With all the elements specified above, we have as prior distribution :

$$\mu \sim \mathcal{N}(\pi, \tau\Sigma)$$

Question c

We impose the view that U.S. equities will outperform global equities by 2% per year. This view is economically defensible based on the fact that over the last decade, U.S. equity markets have consistently benefited from superior productivity growth, technological leadership, and stronger corporate profitability relative to many international markets.
Moreover, in the Black-Litterman framework, the matrix Ω represents the variance of the error associated with the investor's view, and therefore quantifies the degree of confidence placed on that view. A smaller value of Ω corresponds to stronger confidence and gives the view greater influence on the posterior expected returns, while a larger value reflects weaker confidence and places more weight on the market equilibrium prior.

Assuming a normal prior for μ and a normally distributed view error

$$P\mu - Q \sim \mathcal{N}(0, \Omega),$$

With

$$P = \begin{pmatrix} 1 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0.02 \end{pmatrix}, \quad \Omega = \begin{pmatrix} 0.0005 \end{pmatrix}$$

which corresponds to a standard deviation of approximately

$$\sqrt{0.0005} \approx 2.2\%.$$

Bayesian inference yields a posterior distribution that is also multivariate normal:

$$\mu \mid \text{view} \sim \mathcal{N}(\mu_{BL}, \Sigma_{BL}).$$

With

$$\Sigma_{BL} = \Big((\tau\Sigma)^{-1} + P^T\Omega^{-1}P\Big)^{-1}$$

And

$$\mu_{BL} = \Sigma_{BL}\Big((\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}Q\Big) = \begin{pmatrix} 0.0642 \\ 0.0520 \\ -0.0013 \end{pmatrix}$$

Question d

We can compute the mean-variance optimal weights using this formula :

$$w^* = \frac{1}{\delta}\Sigma^{-1}\mu$$

With the view we chose, the posterior allocation is :

$$w_{post} = \begin{pmatrix} 0.71 \\ 0.09 \\ 0.20 \end{pmatrix}$$

We can compare it to the prior allocation, which was:

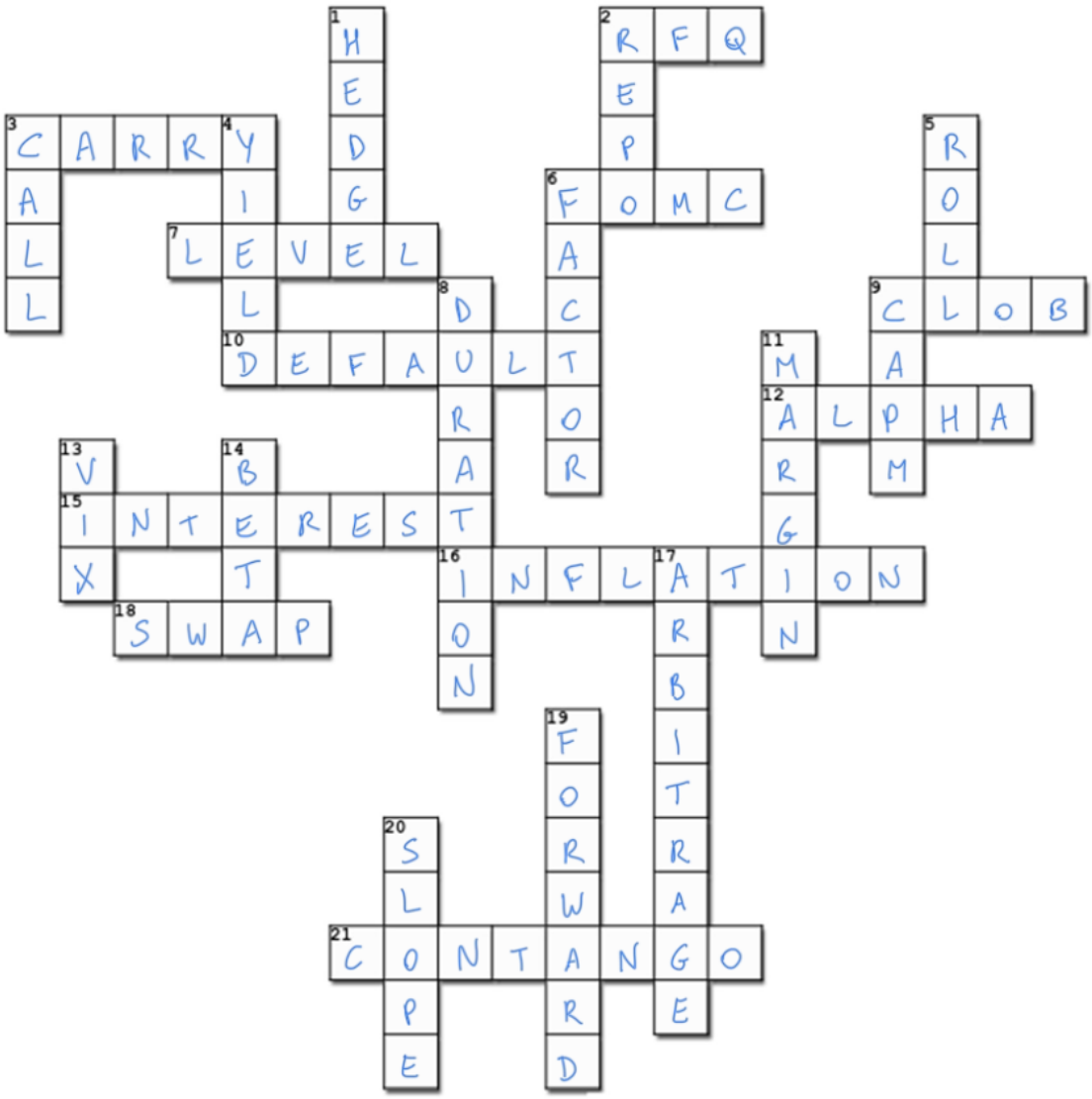
$$w_{prior} = \begin{pmatrix} 0.40 \\ 0.40 \\ 0.20 \end{pmatrix}$$

Then, we can see that the view increases exposure to US equities and decreases allocation to global equities, while Treasuries remain stable. This shift upward for U.S. equities and downward away from global equities in the optimal allocation in the mean–variance optimization problem is coherent with the view we chose that states that U.S. equities are expected to outperform global equities by 2% per year and thus makes U.S. equities more attractive.

```python
import pandas as pd
import numpy as np

## QUESTION A

# Load data
sp500 = pd.read_excel('SPPerformanceGraphExport.xlsx')
spworld = pd.read_excel('SPWorldPerformanceGraphExport.xlsx')
ustreasury = pd.read_excel('USTreasuryPerformanceGraphExport.xlsx')
rf = pd.read_excel('DTB4WK.xlsx')

# Clean index data
def clean_index(df):
    df = df.iloc[:, :2]
    df.columns = ['Date','Value']
    df = df.dropna()
    df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
    df = df.dropna()
    df = df.set_index('Date')
    return df

sp500_c = clean_index(sp500)
spworld_c = clean_index(spworld)
ustreasury_c = clean_index(ustreasury)


## QUESTION B

# Compute returns
rets = pd.concat([
    sp500_c['Value'].pct_change(),
    spworld_c['Value'].pct_change(),
    ustreasury_c['Value'].pct_change()
], axis=1)

rets.columns = ['SP500','SPWorld','USTreasury']
```

```python
# Risk-free rate conversion
rf['observation_date'] = pd.to_datetime(rf['observation_date'])
rf = rf.set_index('observation_date')['DTB4WK']/100/252

rets = rets.dropna()
rf_aligned = rf.reindex(rets.index).fillna(method='ffill')
excess = rets.sub(rf_aligned, axis=0)

# Annualised mean and covariance
mu = excess.mean()*252
Sigma = excess.cov()*252

# Market weights and BL prior
delta = 2.5
tau = 0.05
w_mkt = np.array([60,60,30]) / 150
pi = delta * Sigma.values @ w_mkt


## QUESTION C

# View specification
P = np.array([[1,-1,0]])
Q = np.array([0.02])
Omega = np.array([[0.0005]])

# Posterior calculation
Sigma_bl = np.linalg.inv(np.linalg.inv(tau*Sigma.values) + P.T @ np.linalg.inv(Omega) @ P)
mu_bl = Sigma_bl @ (np.linalg.inv(tau*Sigma.values) @ pi + P.T @ np.linalg.inv(Omega) @ Q)


## QUESTION D

# Portfolio weights prior and posterior
w_prior = (1/delta) * np.linalg.inv(Sigma.values) @ pi
w_BL = (1/delta) * np.linalg.inv(Sigma.values) @ mu_bl

print(mu)
print(Sigma)
print(pi)
print(Sigma_bl)
print(mu_bl)
print(w_prior)
print(w_BL)
```

Question 3



Across

2. A trading mechanism that is essentially an auction
3. A popular strategy in FX market
6. Decision-making body for interest rates in the U.S.
7. A principal component of yields
9. A trading mechanism that aggregates all orders
10. A reason why corporate bonds are risky
12. Excess return (positive or negative) not explained by the model
15. What money pays
16. One of the Federal Reserve's mandate
18. A derivative contract that is not traded on exchange
21. A shape of commodity futures curve

Down

1. One reason to use derivatives
2. How to finance a Treasury security
3. A type of options
4. A measure of interest rates
5. What you have to do for going long in commodity futures
6. A variable that partly explains many stocks' returns
8. How risky a bond is
9. A standard model for calculating expected stock returns
11. Why your broker calls to request money ASAP
13. A popular measure of forward-looking volatility
14. A measure of how much a stock comoves with the market
17. Free money
19. A derivative contract that specifies what happens in the future today
20. A principal component of yields